

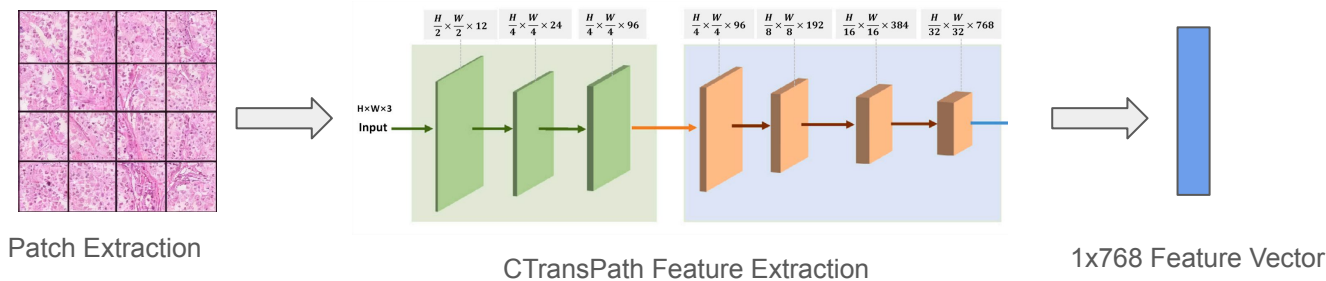


Domain Shift Quantification

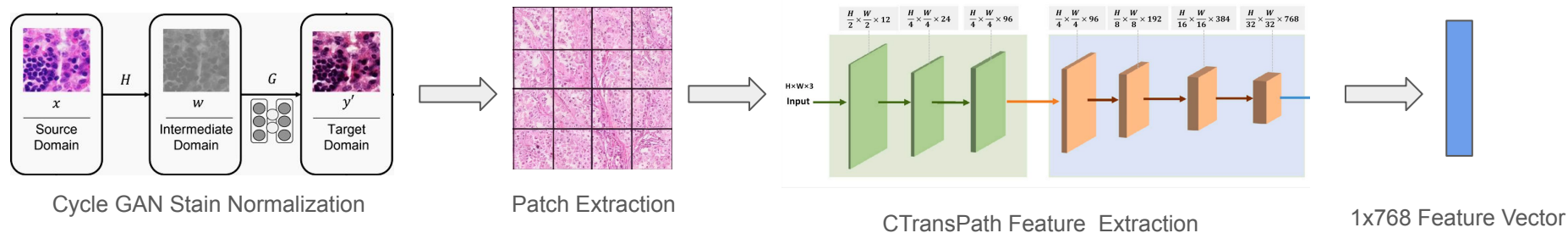
Yang Cao | Lillian Lam | Naina Misra | 02/05/2025

Recap

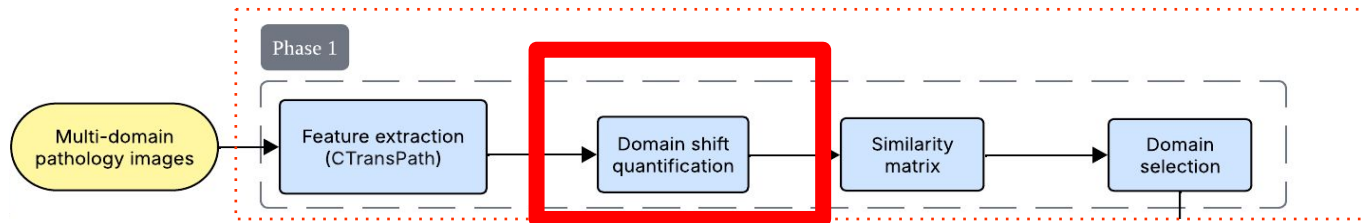
CTransPath Feature Extraction



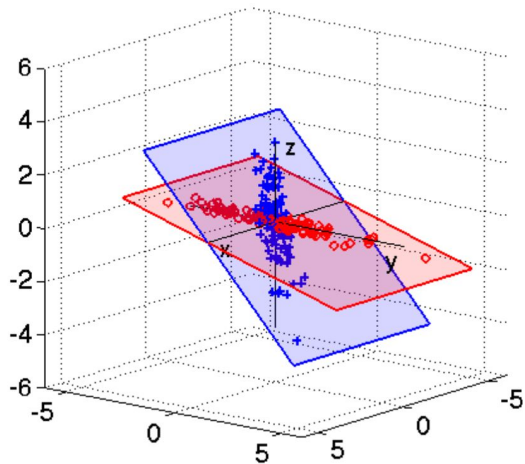
Stain Normalized CTransPath Feature Extraction



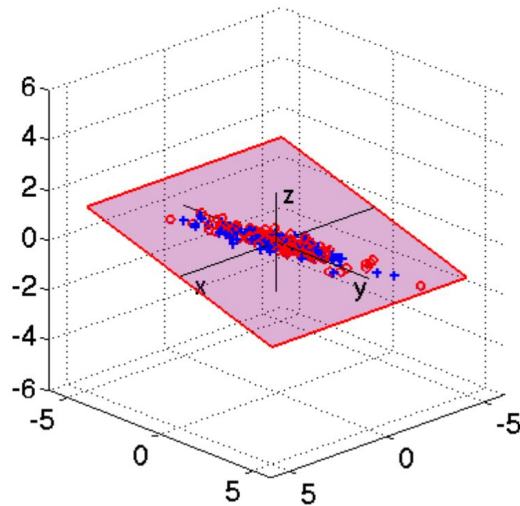
Pipeline



Correlation Alignment (CORAL)



Original Features



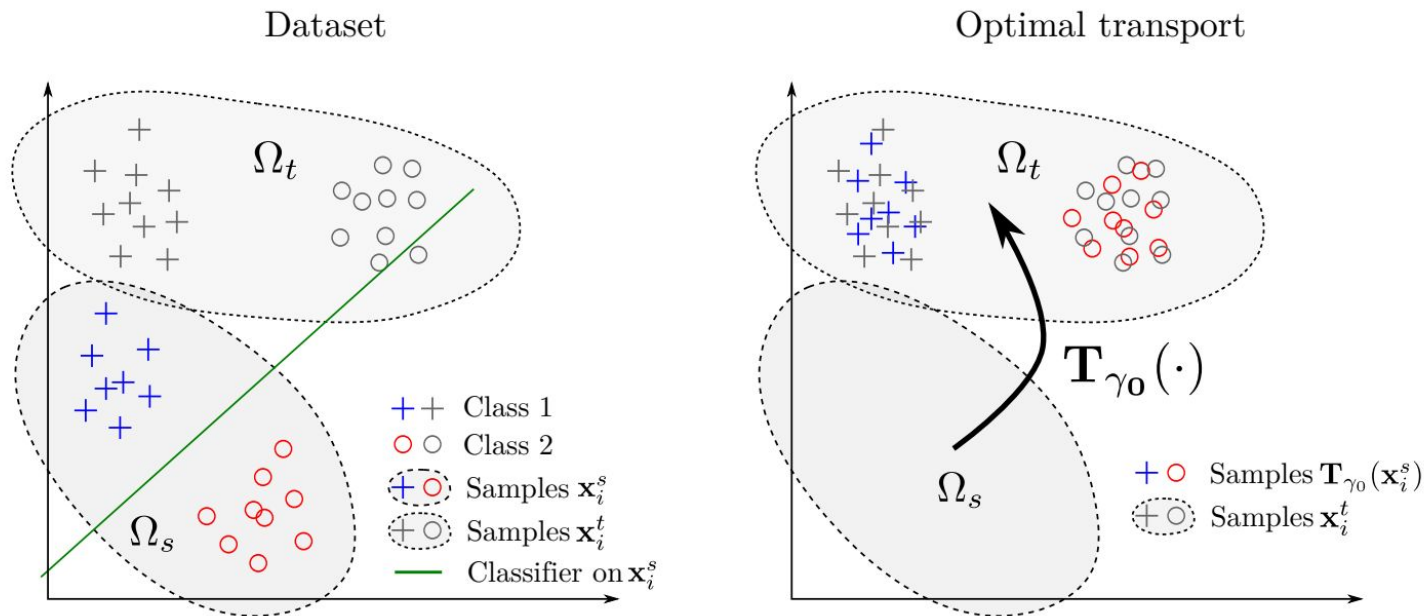
CORAL Aligned

CORAL measures the difference between the Covariance Matrices.

$$L_{CORAL} = \frac{1}{4d^2} \|C_S - C_T\|_F^2$$

where C_S and C_T are the covariance matrices of the source and target domains, and $\|\cdot\|_F$ denotes the Frobenius norm.

Wasserstein Distance



Wasserstein measures the area between the Cumulative Distribution Functions (CDFs).

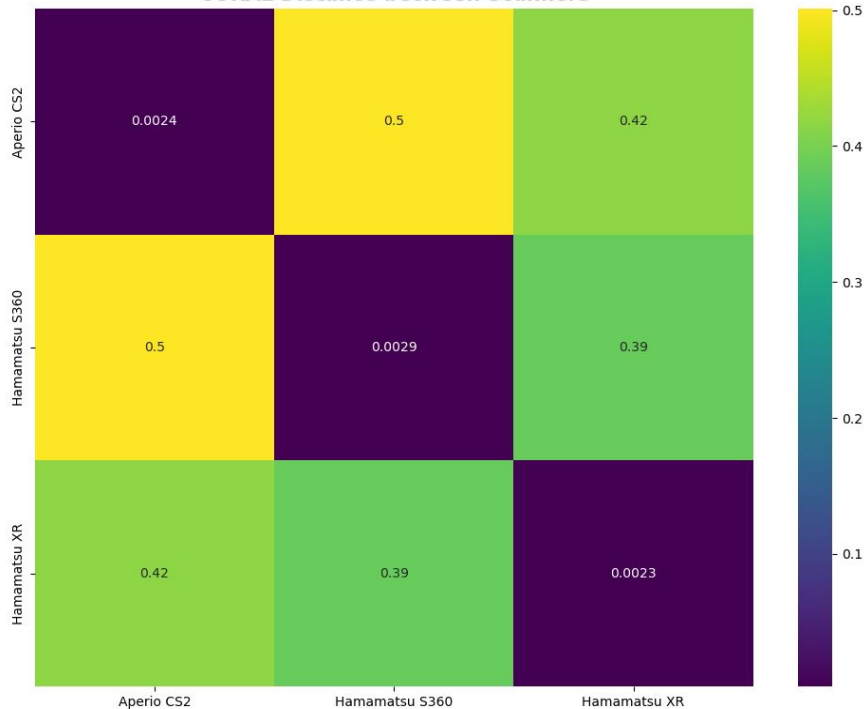
$$W_1(u, v) = \int_{-\infty}^{+\infty} |CDF_u(x) - CDF_v(x)| dx$$

Scanner Differences

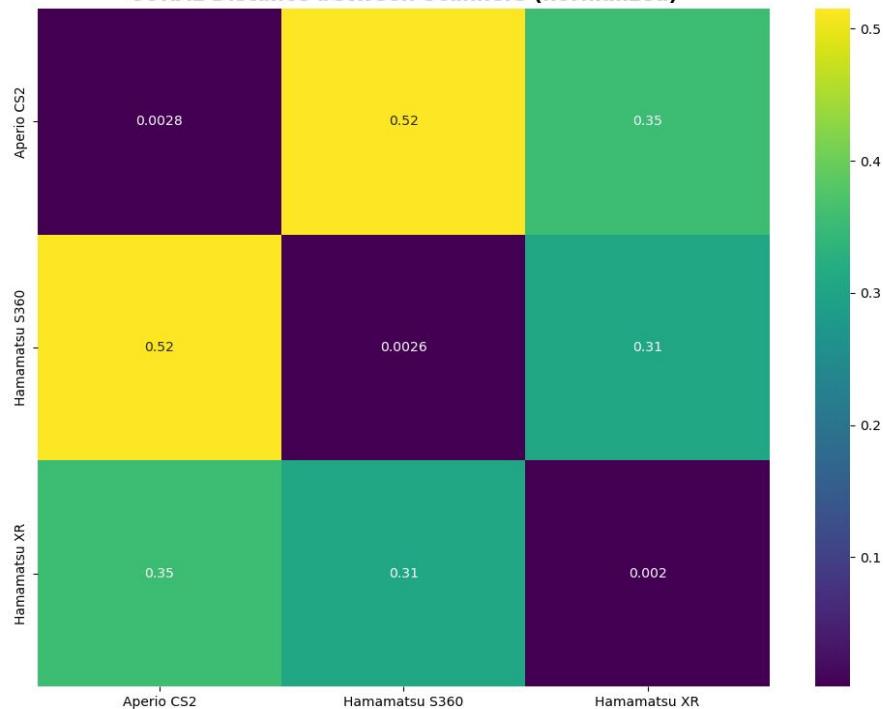
Domain	No. Cases	Tumor Type	Origin	Species	Scanner	Resolution
1a	50	breast carcinoma	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23 \frac{\mu m}{px}$
1b	50	breast carcinoma	UMC Utrecht	human	Hamamatsu S360 (0.5 N/A)	$0.23 \frac{\mu m}{px}$
1c	50	breast carcinoma	UMC Utrecht	human	Leica ScanScope CS2	$0.25 \frac{\mu m}{px}$

Scanner Differences (CORAL)

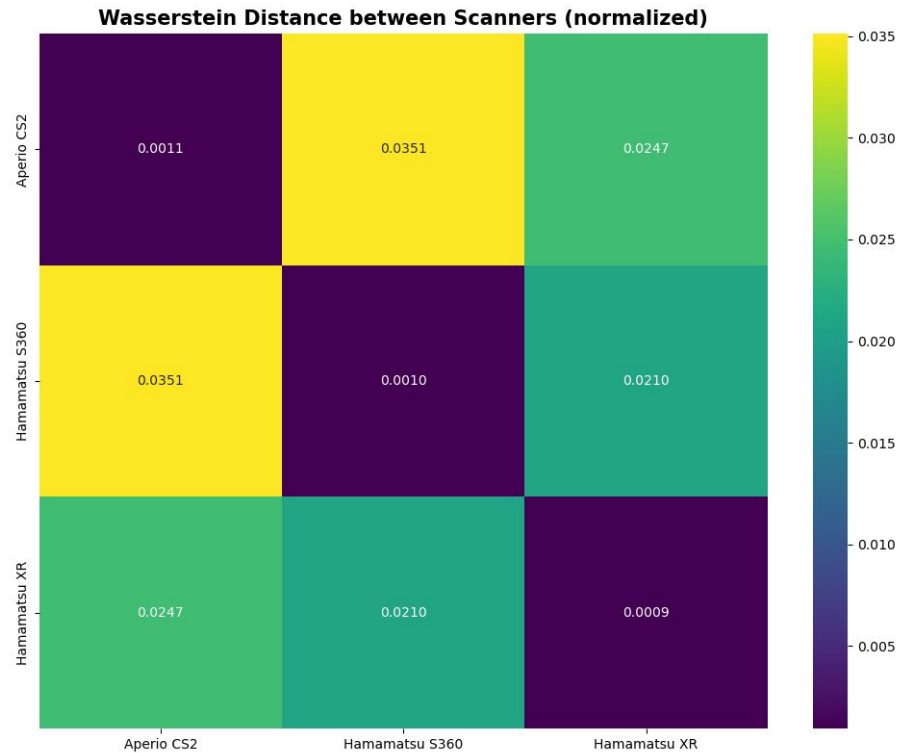
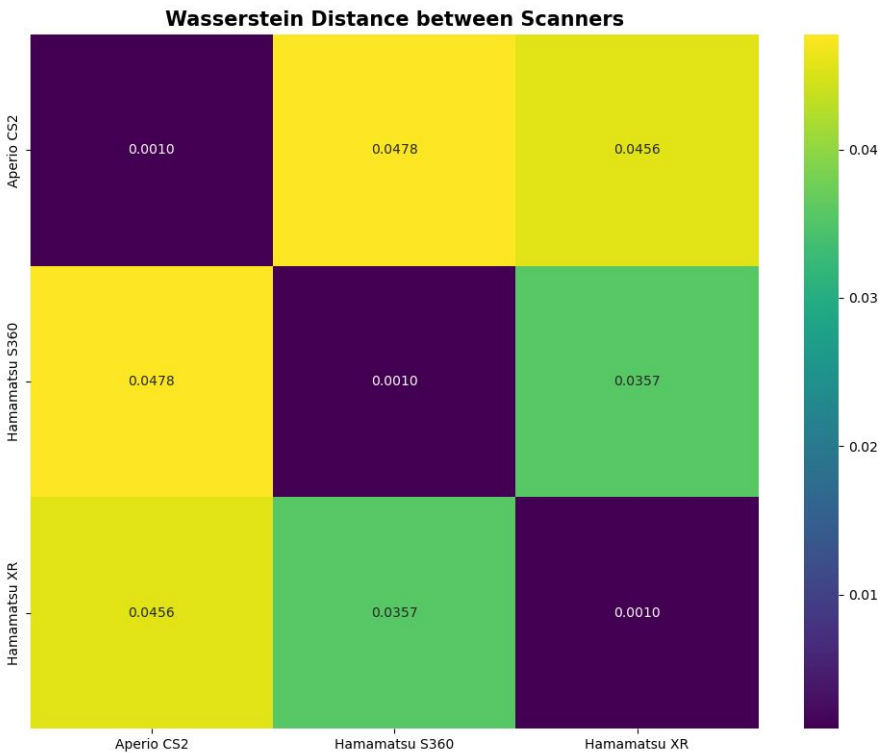
CORAL Distance between Scanners



CORAL Distance between Scanners (normalized)



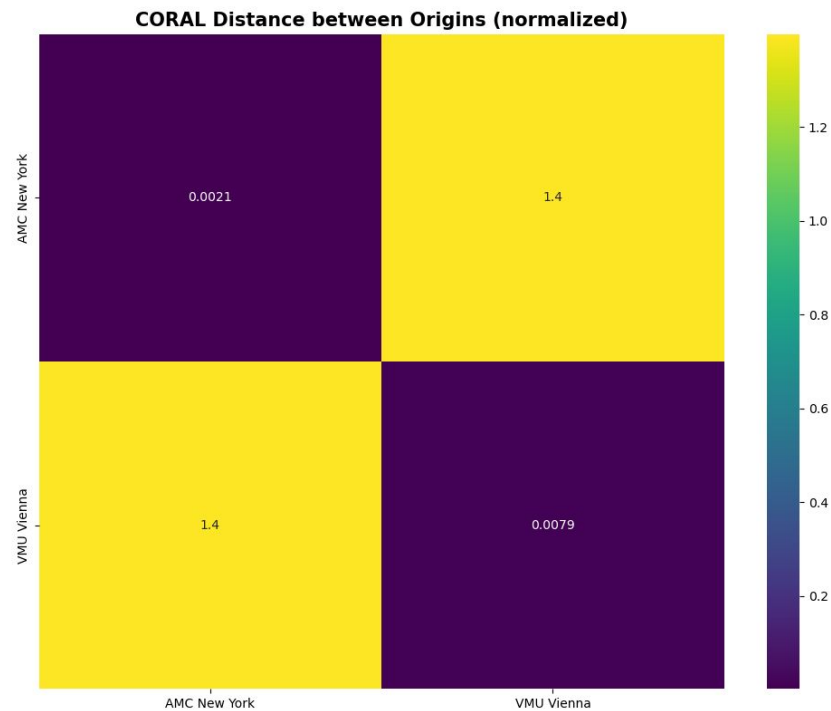
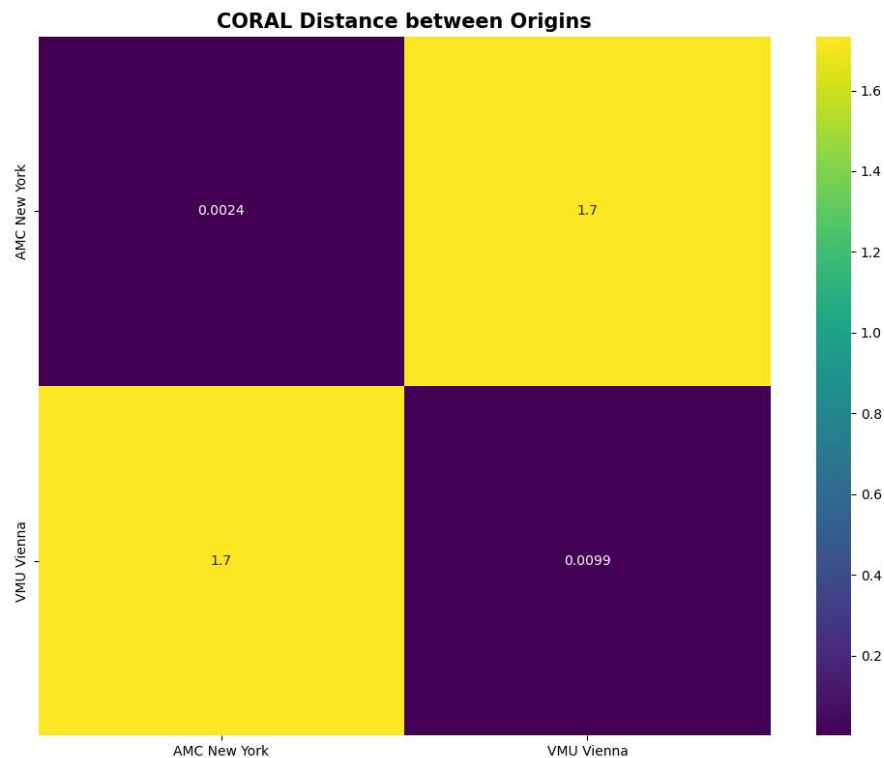
Scanner Differences (Wasserstein)



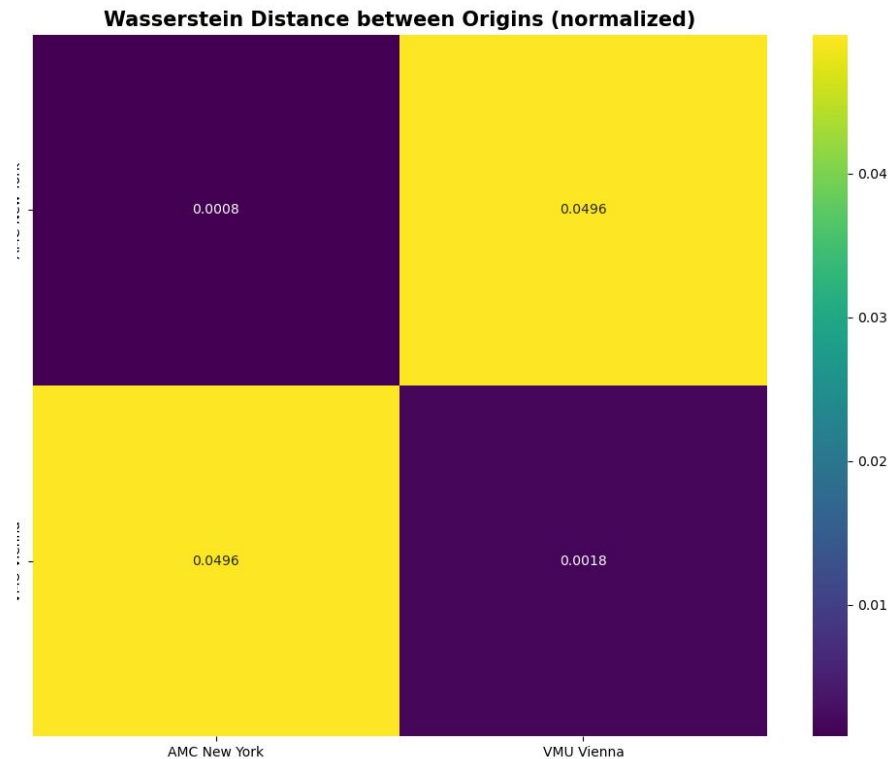
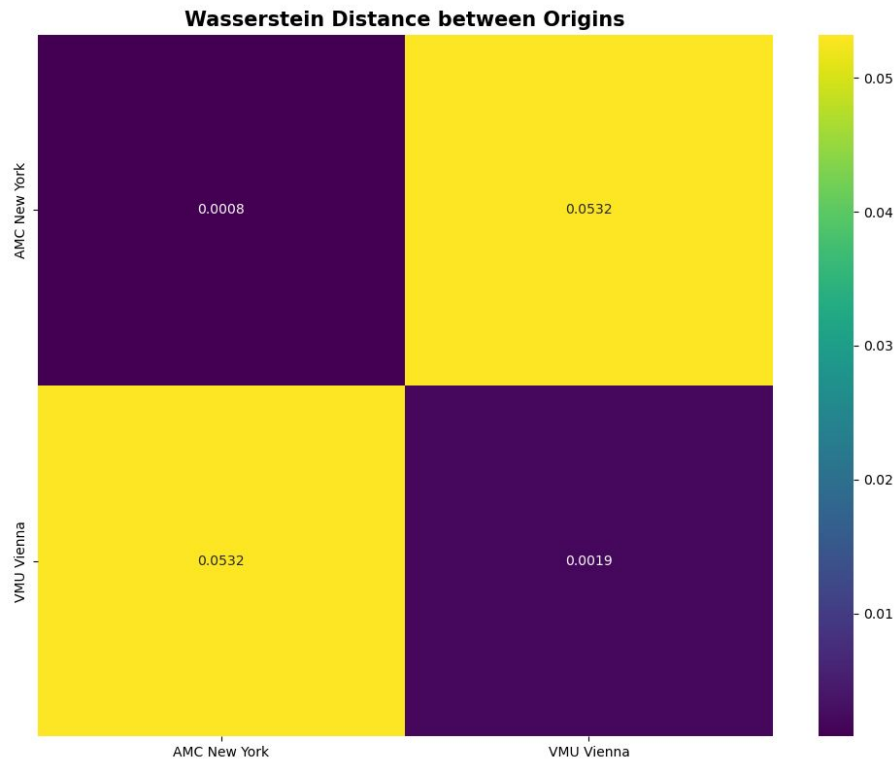
Lab Origin

Domain	No. Cases	Tumor Type	Origin	Species	Scanner	Resolution
6a	85	soft tissue sarcoma	AMC New York	canine	3DHistech Pannoramic Scan II	$0.25 \frac{\mu m}{px}$
6b	15	soft tissue sarcoma	VMU Vienna	canine	3DHistech Pannoramic Scan II	$0.25 \frac{\mu m}{px}$

Lab Origin (CORAL)



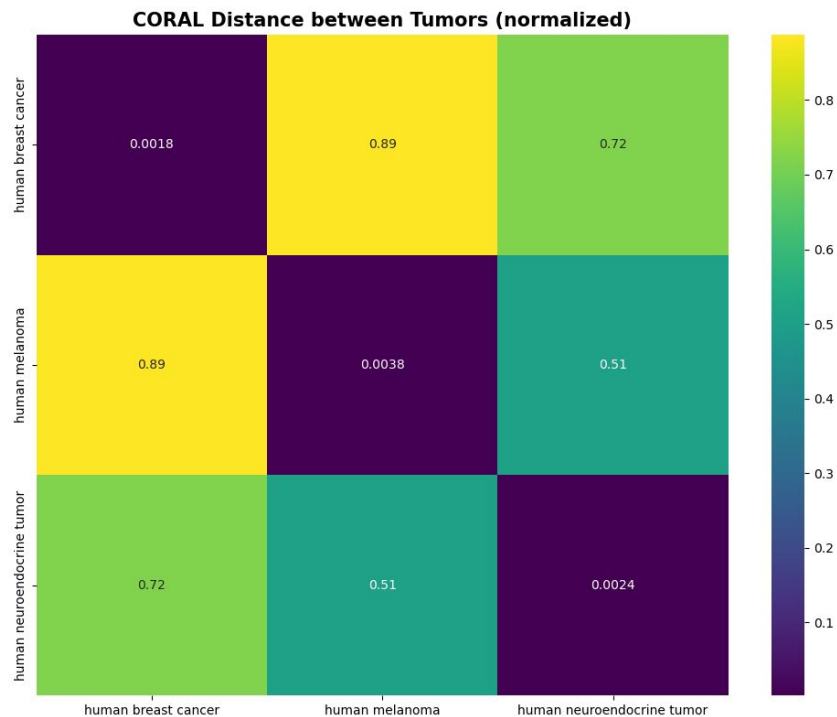
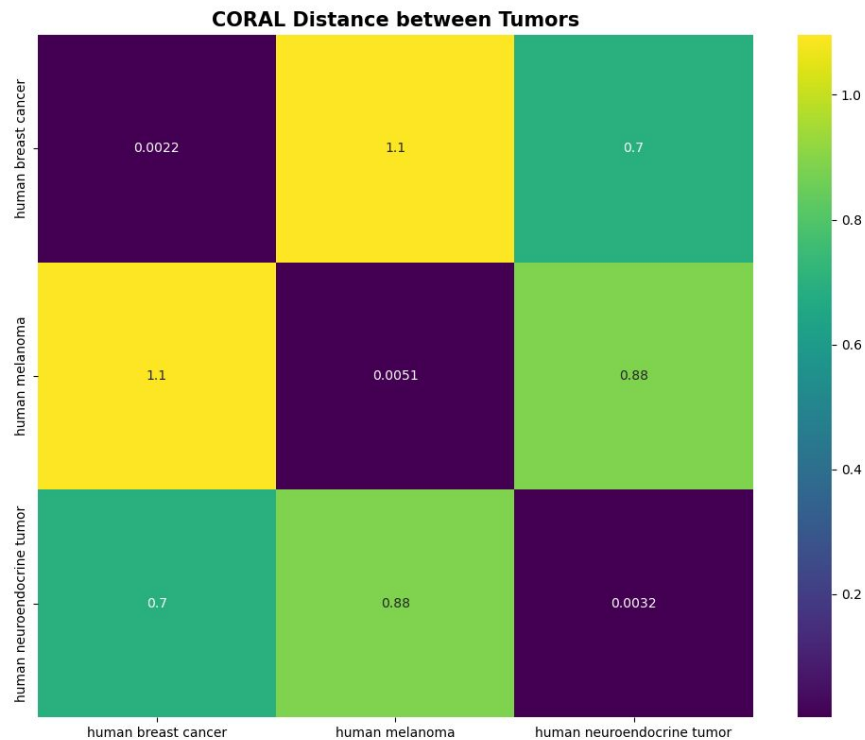
Lab Origin (Wasserstein)



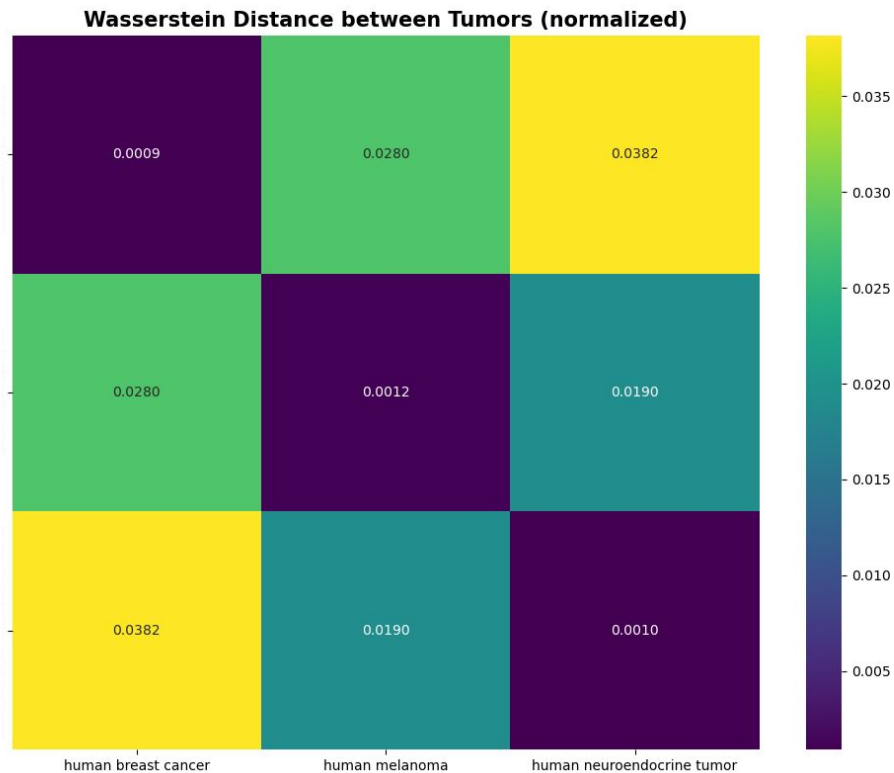
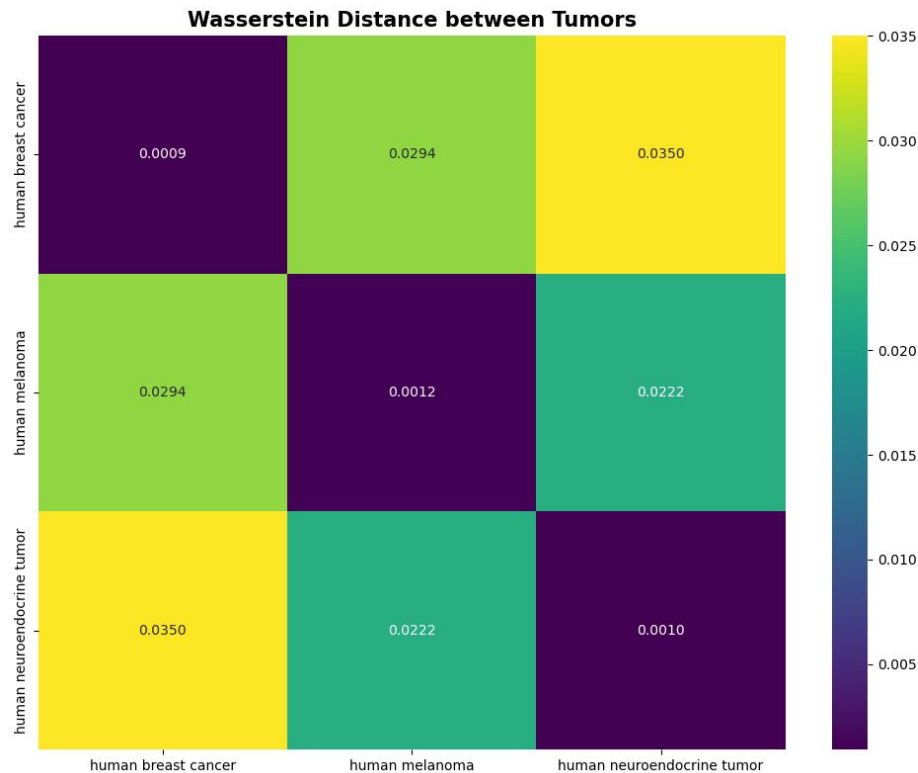
Tumor Differences

Domain	No. Cases	Tumor Type	Origin	Species	Scanner	Resolution
1a	50	breast carcinoma	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23 \frac{\mu m}{px}$
5	55	neuroendocrine tumor	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23 \frac{\mu m}{px}$
7	49	melanoma	UMC Utrecht	human	Hamamatsu XR (C12000-22)	$0.23 \frac{\mu m}{px}$

Tumor Differences (CORAL)



Tumor Differences (Wasserstein)



Summary

- Stain normalization effectively reduced distances between Scanners
- Lab Origin displayed the highest magnitude of shift , confirming that origin encompasses complex variations that normalization alone cannot fully remove

Things Done in Parallel

- Found Ki67 datasets
- started gene prediction from H&E and differential gene expression
- Started training with R-CNN

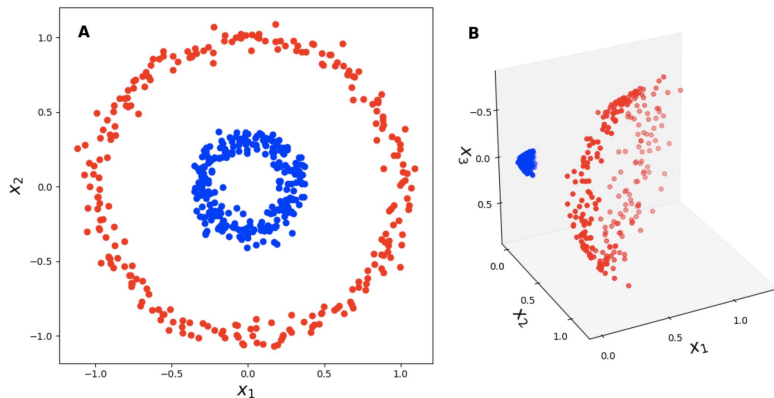
Next Time

- Calculate proxy-A distance for each control
- Formulate the weights for the similarity matrix

Thank you!

Kernel Trick

MMD uses a kernel function to compare distributions in a high-dimensional space where differences are easier to see.

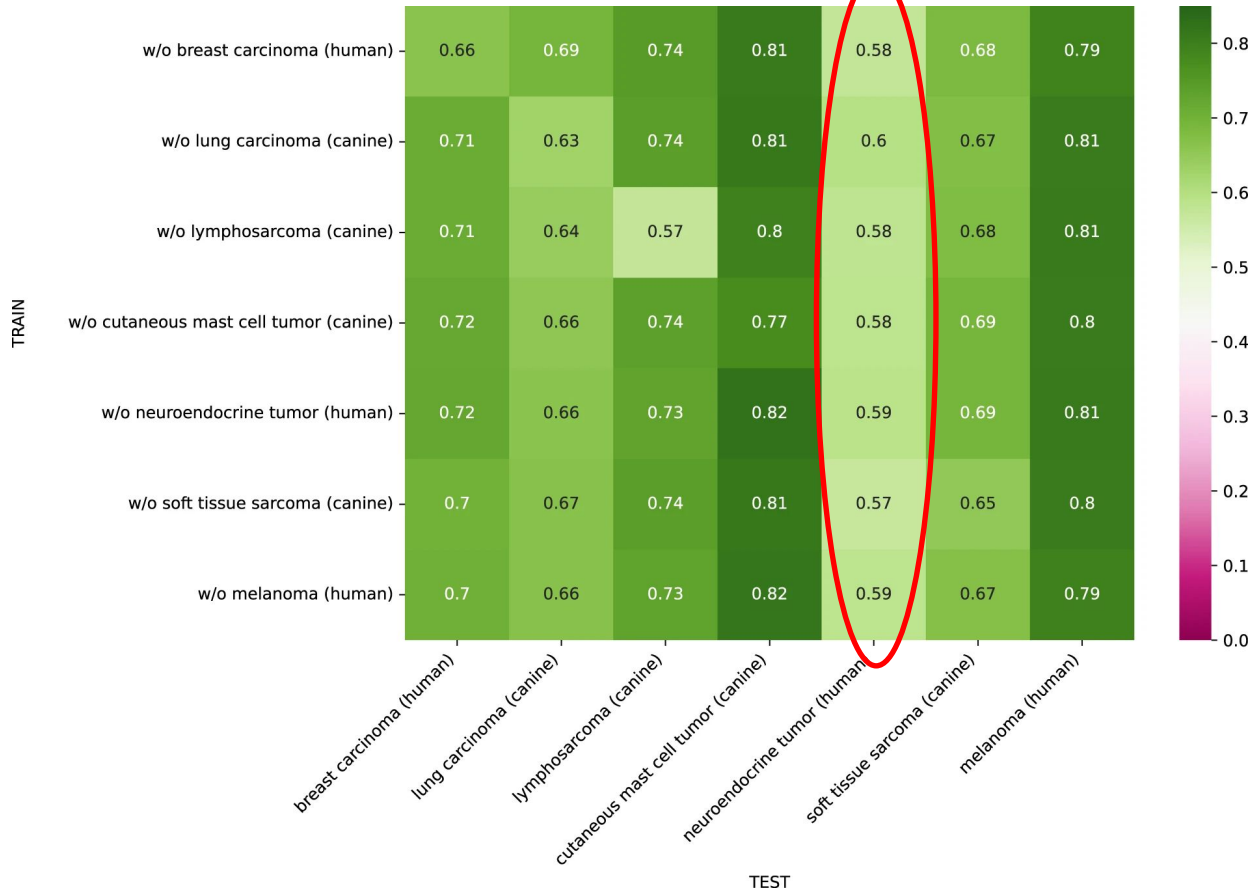


Kernel Choice: Linear Kernel (basic dot product)

$$k(x, y) = x_1y_1 + x_2y_2 + \dots + x_ny_n$$

- CTransPath features already high-dimensional (768D)
- Computationally efficient - no hyperparameters (gaussian kernel requires γ)
- Interpretable - measures similarity in original space
- Stable - no gamma parameter tuning needed

Leave-One-Domain-Out Training from Paper



Maximum Mean Discrepancy (MMD)



Set X: 100 images of NYC



Set Y: 100 images of HK

MMD measures: How different are these collections on average?

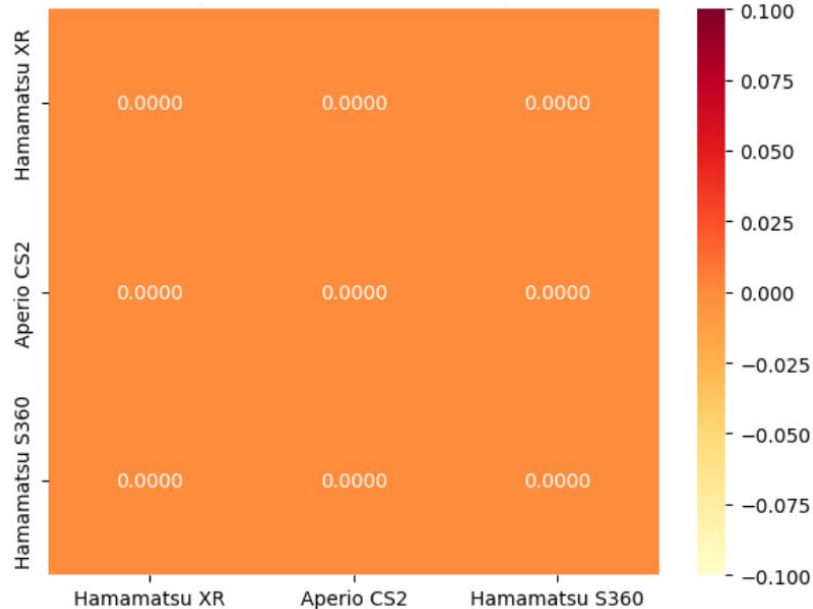
$$\text{MMD}^2 = \frac{1}{n(n-1)} \sum_i \sum_{j \neq i} k(x_i, x_j) + \frac{1}{m(m-1)} \sum_i \sum_{j \neq i} k(y_i, y_j) - \frac{2}{nm} \sum_i \sum_j k(x_i, y_j)$$

Problem: Features have small means

Scanner Hamamatsu XR:
Samples: (29952, 768)
Feature mean: 0.0005
Feature std: 0.1338
Any NaN: False

Scanner Aperio CS2:
Samples: (23142, 768)
Feature mean: 0.0008
Feature std: 0.1341
Any NaN: False

Scanner Hamamatsu S360:
Samples: (28015, 768)
Feature mean: 0.0005
Feature std: 0.1380
Any NaN: False



Dot product of small number $\Rightarrow$ small results (floating-point precision limits)

Solution: Feature Scaling (divide features by pooled standard deviation)